

Basic Electrical Engineering

MODULE 3 — PART 1: COMPLETE NOTES

Module 3 (6 hrs): Basics of Magnetic Circuits and DC Machines

For First-Year Students — written in simple, easy language

Course: Basic Electrical Engineering · Course Code: EE100

Module 3 — what it covers (as per the syllabus)

- Basics of the magnetic circuit
- MMF, flux and reluctance calculations for simple magnetic cores
- B-H curve
- DC machines: construction, principle of operation, basic equations and applications

Contents

Section	Topic
1	Magnetic Field and Magnetic Force
2	Magnetic Flux and Flux Density
3	Electromagnet, Conductor and Coil
4	Magnetomotive Force (MMF)
5	Magnetic Field Strength
6	Permeability
7	Reluctance
8	Composite Series Magnetic Circuits
9	B-H Curve and Hysteresis Loop
10	Electromagnetic Induction
11	DC Machines — Introduction and Classification
12	DC Generator — Principle of Operation
13	Construction of DC Machines
14	EMF Equation of a DC Generator
15	Types of DC Generator and Applications
16	Force on a Current-Carrying Conductor
17	DC Motor — Principle, Torque and Back EMF
18	Solved Numerical Examples
19	Quick Reference — Formulae and Definitions

1. Magnetic Field and Magnetic Force

The space around the poles of a magnet is called the **magnetic field**. It is represented by magnetic lines of force. The force exerted by one magnet on another, either of attraction or of repulsion, is called the **magnetic force**. According to Coulomb's first law, **unlike poles attract** each other and **like poles repel** each other.

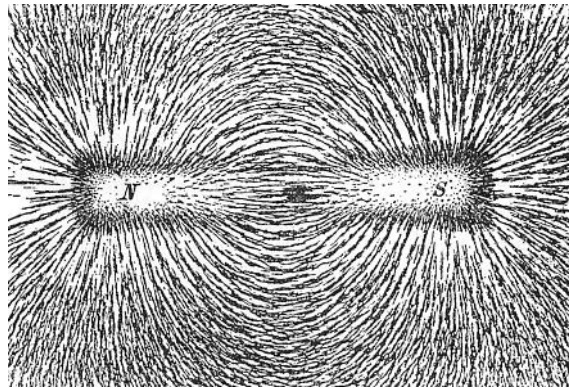


Figure 1: The magnetic field around a bar magnet.

2. Magnetic Flux and Flux Density

2.1 Magnetic Flux (ϕ)

The number of magnetic lines of force set up in a magnetic circuit is called the **magnetic flux**. It is analogous to the electric current in an electric circuit. Its unit is the **weber (Wb)**. Its important properties are:

- It always forms a closed loop.
- It always starts from the north pole and ends at the south pole (outside the magnet).
- Lines of flux never intersect each other.

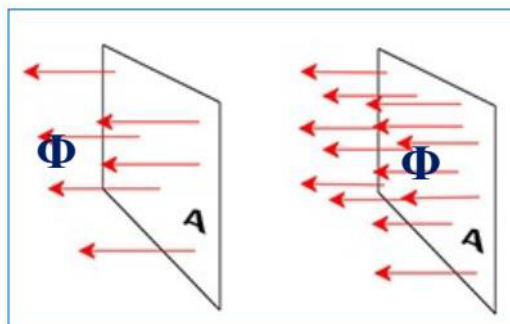


Figure 2: Magnetic flux lines forming closed loops.

2.2 Magnetic Flux Density (B)

The magnetic flux density is the amount of flux passing through an area perpendicular to the direction of the flux. Its unit is the **tesla (T)** or Wb/m^2 .

$$B = \phi / A \quad (\text{tesla})$$

Example: a pole face of 200 mm × 100 mm has a total flux of 150 μWb . Then $A = 0.2 \times 0.1 = 0.02 \text{ m}^2$ and $B = 150 \times 10^{-6} / 0.02 = 7.5 \text{ mT}$.

3. Electromagnet, Conductor and Coil

3.1 Electromagnet

An electromagnet is a magnet in which the magnetic field is produced by an electric current. Electromagnets usually consist of wire wound into a coil. The main advantage over a permanent magnet is that the field can be changed quickly by controlling the current, but it needs a continuous supply of current to maintain the field.

3.2 Magnetic Field of a Current-Carrying Conductor

A current-carrying conductor creates a magnetic field around it. For a straight conductor the field lines are concentric circles centred on the axis of the conductor. The density of the field is directly proportional to the current, and the direction of the circular field lines is given by **Maxwell's right-hand grip rule**.

Right-hand grip rule: hold the conductor in the right hand so that the fingers wrap around it and the thumb is stretched along the current. The wrapped fingers then show the direction of the circular magnetic field lines.

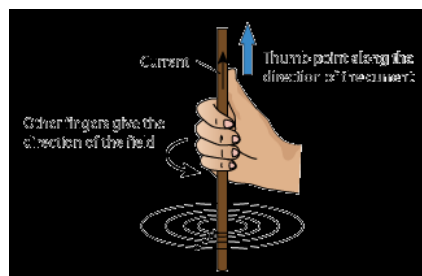


Figure 3: Maxwell's right-hand grip rule.

3.3 Solenoid (Coil)

A current-carrying wire wound spirally in the form of a helix about an axis is called a **solenoid** or coil. It acts just like a bar magnet having north and south poles.

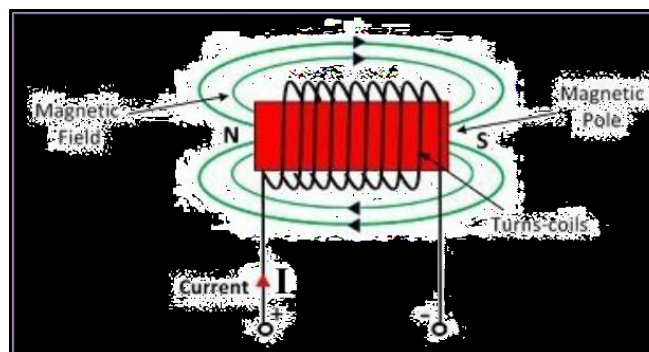


Figure 4: A solenoid acts like a bar magnet.

4. Magnetomotive Force (MMF)

The magnetomotive force is the cause of the existence of magnetic flux in a magnetic circuit. It is the magnetic analogue of the electromotive force (emf) in an electric circuit. Its unit is the **ampere-turn (AT)**.

$$F = N I \quad (\text{ampere-turns})$$

where N is the number of turns and I is the current through the conductor in amperes.

$$F_m = N \times I \text{ (AT).}$$

Figure 5: Magnetomotive force of a coil.

5. Magnetic Field Strength (Magnetising Force)

The magnetic field strength is defined as the magnetomotive force per unit length of the flux path. Its unit is **ampere-turns per metre (AT/m)**.

$$H = F / l = N I / l \quad (\text{AT/m})$$

where l is the mean length of the flux path in metres.

$$H = \frac{mmf}{l}$$

Figure 6: Magnetic field strength (magnetising force).

6. Permeability

For air or any non-magnetic medium, the ratio of flux density B to magnetising force H is constant. This constant is called the **permeability of free space**, $\mu_0 = 4\pi \times 10^{-7}$ henry per metre. For any other medium the absolute permeability is:

$$\mu = \mu_0 \mu_r \quad \text{and} \quad B = \mu H$$

where μ_r is the **relative permeability** of the medium ($\mu_r = 1$ for air, and much greater than 1 for iron and steel).

Example: for an air-gap of length 12 mm and flux density 0.25 T, $H = B/\mu_0 = 0.25/(4\pi \times 10^{-7}) = 198944 \text{ AT/m}$, so the mmf = $H l = 198944 \times 0.012 = 2387 \text{ AT}$.

7. Reluctance

The resistance offered by a magnetic circuit to the presence of magnetic flux is called **reluctance**. It is the magnetic analogue of electrical resistance. Its unit is **AT/Wb**.

$$S = l / (\mu_0 \mu_r A) = l / (\mu A) \quad (\text{AT/Wb})$$

where l is the length of the magnetic path, A is the cross-sectional area, and μ is the permeability of the material.

$$\text{Reluctance } S = \frac{l}{\mu_0 \mu_r A} = \frac{150 \times 10^{-3}}{(4\pi \times 10^{-7})(4000)(1800 \times 10^{-6})} \\ = 16\,580/\text{H}$$

Figure 7: Reluctance of a magnetic circuit.

Analogy between the magnetic and electric circuits

Magnetic circuit	Electric circuit
Magnetomotive force F (AT)	Electromotive force E (V)
Flux ϕ (Wb)	Current I (A)
Reluctance S (AT/Wb)	Resistance R (Ω)
Permeability μ	Conductivity σ

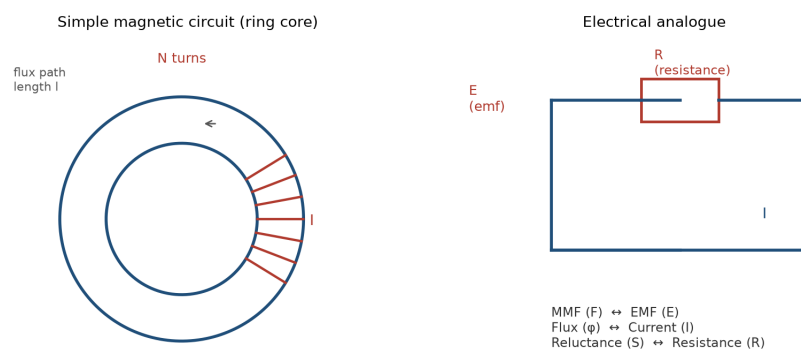


Figure 8: A simple magnetic circuit and its electrical analogue.

8. Composite Series Magnetic Circuits

A composite series magnetic circuit has two or more parts in series (for example iron and an air-gap, or two different lengths of iron). The same flux passes through every part, and the total reluctance is the sum of the reluctances of all the parts:

$$S = S_1 + S_2 + \dots + S_n$$

The total mmf required is then $F = \phi S$. This is used to find the current needed in the coil to produce a required flux density in an air-gap.

$$l = 2\pi r = 2 \times \pi \times 50 \times 10^{-3} \text{ m}; A = 400 \times 10^{-6} \text{ m}^2; I = 0.5 \text{ A}; \\ \Phi = 0.1 \times 10^{-3} \text{ Wb}; \mu_r = 200$$

$$(a) \text{ Reluctance } S = \frac{l}{\mu_0 \mu_r A} = \frac{2 \times \pi \times 50 \times 10^{-3}}{(4\pi \times 10^{-7})(200)(400 \times 10^{-6})} \\ = 3.125 \times 10^6 / \text{H}$$

Figure 9: A composite series magnetic circuit.

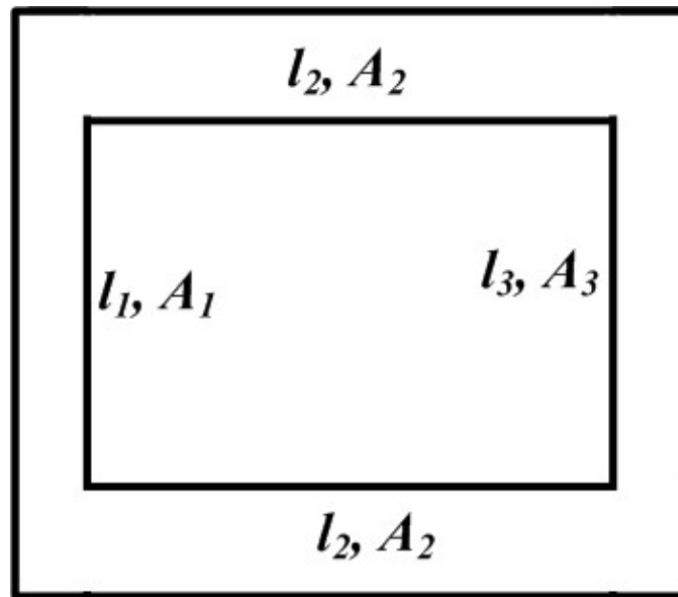


Figure 10: A magnetic circuit with an air-gap.

9. B-H Curve and Hysteresis Loop

The **B-H curve** (magnetisation curve) shows how the flux density B in a magnetic material varies with the magnetising force H . It has three parts:

- A **linear (unsaturated)** region at low H , where B is proportional to H .
- A **knee point**, where the curve begins to bend.
- A **saturation** region, where a further increase in H produces very little increase in B .

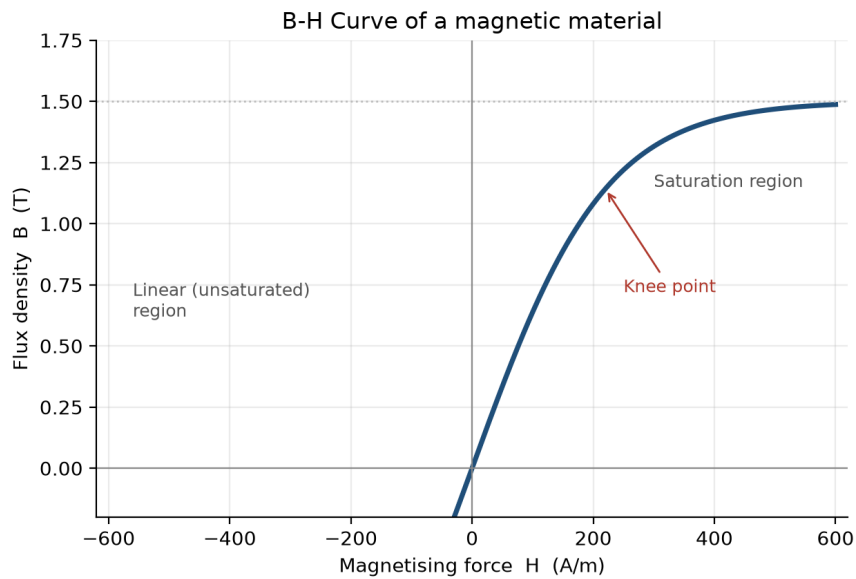


Figure 11: The B-H curve of a magnetic material.

If the magnetising force is taken through a complete cycle of increasing and decreasing values, B does not retrace the same path. The loop so formed is called the **hysteresis loop**.

- **Retentivity (B_r)** is the flux density that remains when H is reduced to zero.
- **Coercivity (H_c)** is the reverse magnetising force needed to reduce B to zero.
- The area of the loop represents the energy lost as heat per cycle (hysteresis loss).

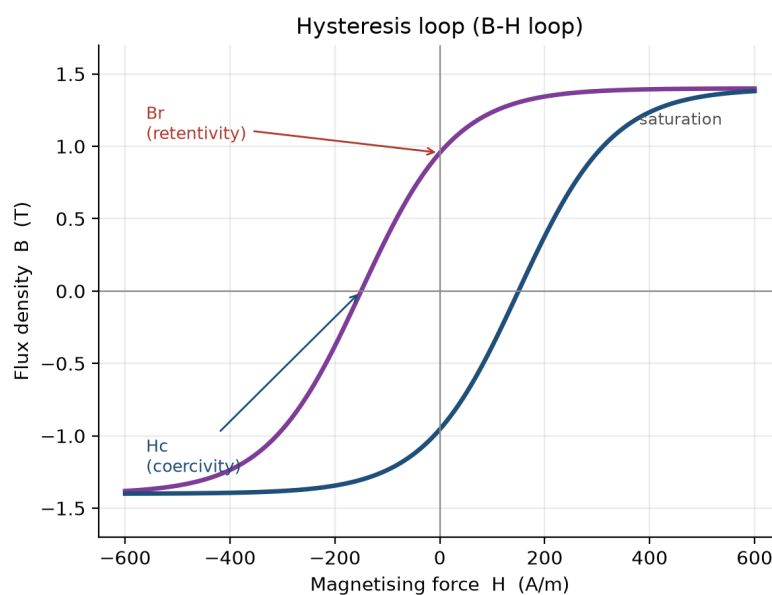


Figure 12: The hysteresis loop, showing retentivity and coercivity.

10. Electromagnetic Induction

The electromagnetic system is an essential element of all rotating electrical machines, electromechanical devices and static devices such as the transformer. An **electric generator** converts mechanical energy into electrical energy and works on the principle of electromagnetic induction; a **motor** converts electrical energy into mechanical energy.

10.1 Faraday's Law and Lenz's Law

When the flux linking an electric circuit changes, an emf is induced whose magnitude is directly proportional to the rate of change of flux linkages (Faraday's law):

$$e = - N (d\phi / dt)$$

The negative sign shows **Lenz's law**: the induced emf is in such a direction that the current it produces sets up a magnetic field opposing the change of flux.

10.2 Dynamically Induced EMF

When a conductor moves in a uniform magnetic field, the emf induced in this way is called a **dynamically induced emf**. Its direction is given by **Fleming's right-hand rule**, and its magnitude is:

$$e = B l v \sin \theta$$

where B is the flux density, l the active length of the conductor, v the relative velocity, and θ the angle between the plane of rotation and the plane of the magnetic flux.

Fleming's right-hand rule

Stretch the thumb, forefinger and middle finger of the right hand mutually perpendicular. If the thumb gives the direction of motion of the conductor and the forefinger the direction of the magnetic field, then the middle finger gives the direction of the induced current.

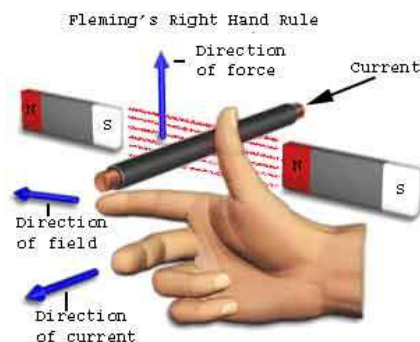


Figure 13: Fleming's hand rule (right hand for generator action).

10.3 Statically Induced EMF

When the conductor and the field are stationary but the magnitude of the linking flux is changed by changing the current, the emf induced is called a **statically induced emf**. It is of two kinds:

- **Self-induced emf** — produced in a coil when the current through its own winding changes.
- **Mutually induced emf** — produced in coil 2 when the current in coil 1 (and hence the flux linking coil 2) changes.

11. DC Machines — Introduction and Classification

A DC machine is a device that converts mechanical energy into electrical energy or vice versa. DC machines are of two types:

- **DC generator** — converts mechanical energy into electrical energy.
- **DC motor** — converts electrical energy into mechanical energy.

Construction-wise there is no basic difference between a DC generator and a DC motor; the same machine can work as either.

12. DC Generator — Principle of Operation

A DC generator is based on the principle that whenever flux is cut by a conductor, an emf is induced, which causes a current to flow if the conductor circuit is closed. Its essential parts are a magnetic field (the stationary stator) and a conductor or armature (the rotating rotor). In a DC generator the field current is DC, producing a constant field flux, so the relative motion is obtained by rotating the conductors in a constant field.

The emf generated in the loop is alternating. It is converted into direct voltage by a device called the **commutator**, which is a mechanical rectifier. The commutator reverses the connection of the coil to the external load at the same instant that the current in the coil reverses, so the current through the load is direct current.

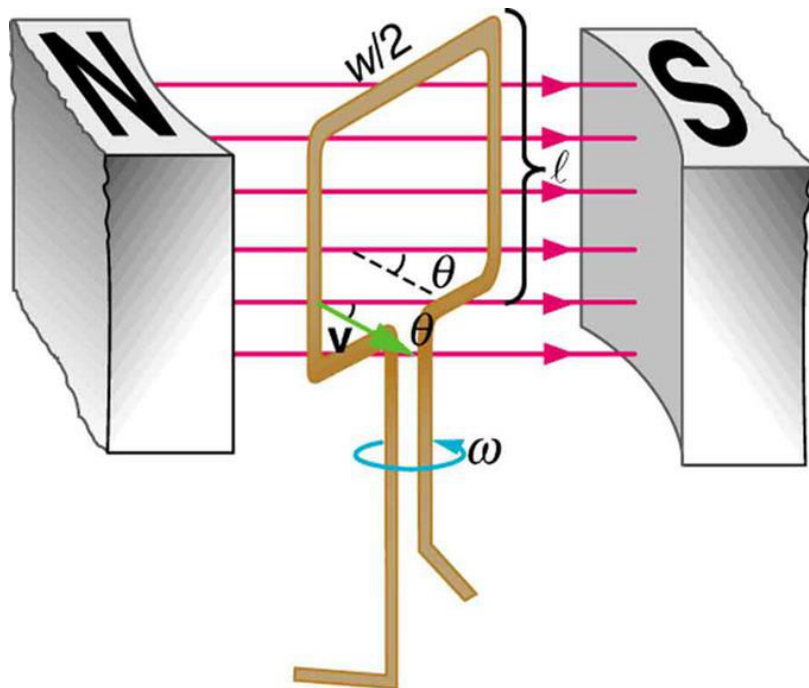


Figure 14: Working principle of a DC generator.

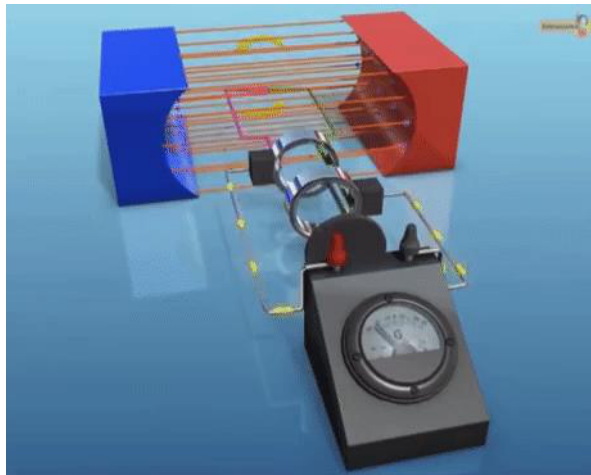


Figure 15: Generation of emf in a rotating loop.

13. Construction of DC Machines

A DC machine has the following main components:

Component	Function
Magnetic frame (yoke)	Outer frame; protects the machine, supports the poles and carries the magnetic flux. Made of cast iron (small machines) or cast/rolled steel (large machines).
Pole core and pole shoe	Fitted on the inner wall of the yoke; the pole shoe spreads the flux over the air-gap to reduce reluctance.
Field winding	Copper coils wound on the pole cores; when the DC field current flows, adjacent poles get opposite polarity and a field flux is produced. It is a stationary part.
Armature core	Houses the armature conductors and gives a low-reluctance path to the flux; keyed to the shaft, drum-shaped, built of laminations to reduce eddy-current loss.
Armature winding	Placed in the armature slots; this is the winding in which the emf is induced. The rotating part.
Commutator	A mechanical rectifier that converts the AC in the armature into unidirectional current; wedge-shaped segments insulated by mica.
Brushes and bearings	Carbon/graphite blocks that collect current from the commutator; bearings support the shaft.

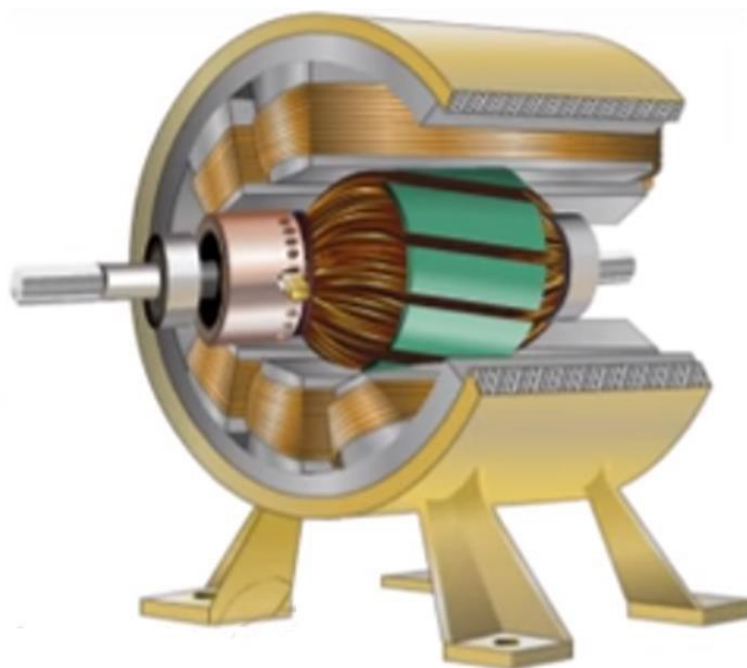


Figure 16: Construction of a DC machine.

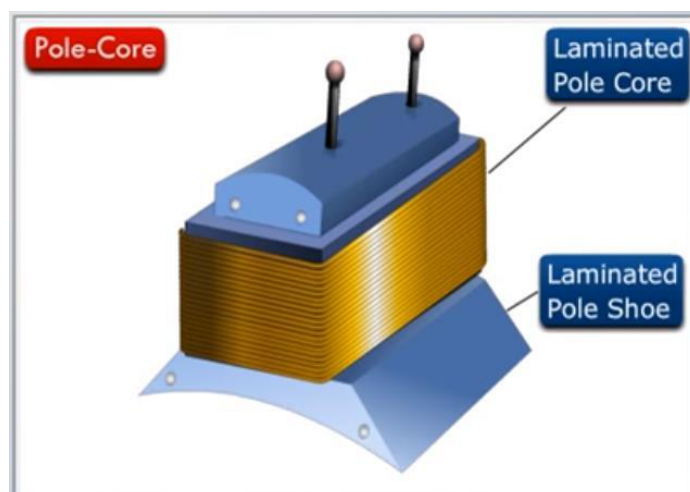


Figure 17: Pole core, pole shoe and field winding.

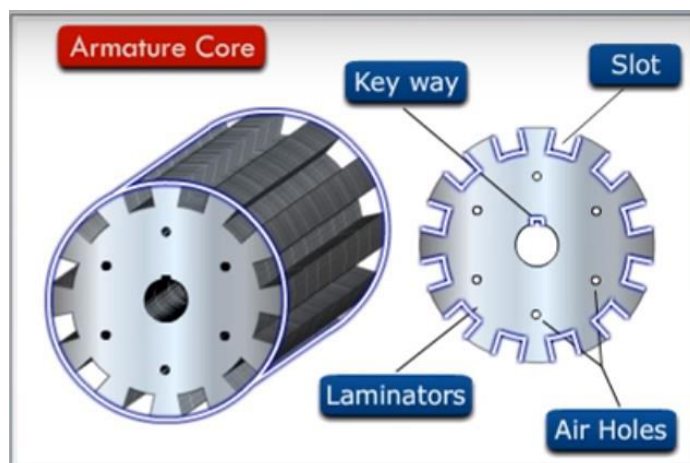


Figure 18: Armature core and armature winding in slots.

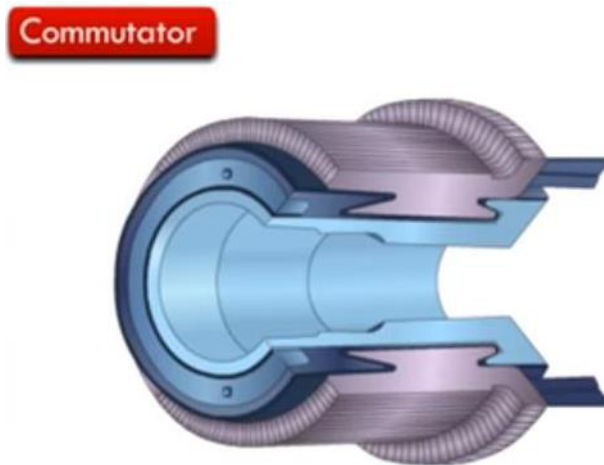


Figure 19: The commutator and its segments.



Figure 20: Cross-sectional view of a DC machine.

13.1 Lap and Wave Windings

Winding	Number of parallel paths	Suitable for
Lap winding	$A = P$ (number of poles)	High-current, low-voltage machines (e.g. welding plant)
Wave winding	$A = 2$ (always)	High-voltage, low-current machines (e.g. lighting generators)

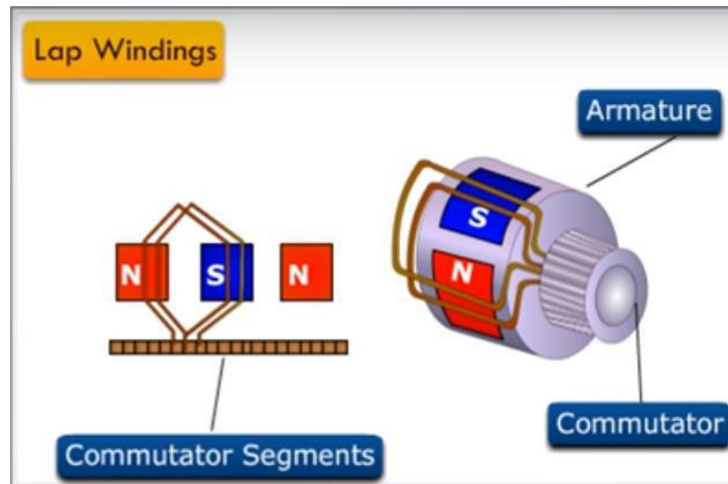


Figure 21: Lap and wave armature windings.

14. EMF Equation of a DC Generator

The emf generated in a DC generator is the resultant emf per parallel path, which equals the average emf per conductor multiplied by the number of conductors in series per parallel path:

$$E = (\phi Z N P) / (60 A) \text{ volts}$$

where:

- ϕ = flux per pole in weber (Wb)
- Z = total number of armature conductors
- N = speed of the armature in revolutions per minute (rpm)
- P = number of poles
- A = number of parallel paths in the armature ($A = P$ for lap, $A = 2$ for wave)

= No. of slots $\times$ No. of conductors/slot

P = No. of generator poles

A = No. of parallel paths in armature

N = armature rotation in revolutions per minute (r.p.m.)

E = e.m.f. induced in any parallel path in armature

Generated e.m.f. E_g = e.m.f. generated in any one of the parallel paths i.e. E .

Average e.m.f. generated/conductor = $\frac{d\Phi}{dt}$ volt ($\because n = 1$)

Now, flux cut/conductor in one revolution $d\Phi = \Phi P \text{ Wb}$

No. of revolutions/second = $N/60$ $\therefore$ Time for one revolution, $dt = 60/N$ second

Hence, according to Faraday's Laws of Electromagnetic Induction,

$$\text{E.M.F. generated/conductor} = \frac{d\Phi}{dt} = \frac{\Phi P N}{60} \text{ volt}$$

Figure 22: Derivation of the EMF equation.

Example: a 4-pole generator has $Z = 500$, $\phi = 0.02 \text{ Wb}$ and $N = 1200 \text{ rpm}$. For a lap winding ($A = 4$), $E = (0.02 \times 500 \times 1200 \times 4)/(60 \times 4) = 200 \text{ V}$.

15. Types of DC Generator and their Applications

Depending on how the field winding is connected, DC generators are classified as:

Type	Application
Separately excited generator	Electroplating, electro-refining of materials (needs a separate field supply).
Series generator	Boosters on DC feeders, constant-current generator for welding and arc lamps.
Cumulative compound generator	Domestic lighting and transmission of energy over long distances.
Differential compound generator	Electric arc and welding.
Shunt generator	General lighting and power supply.

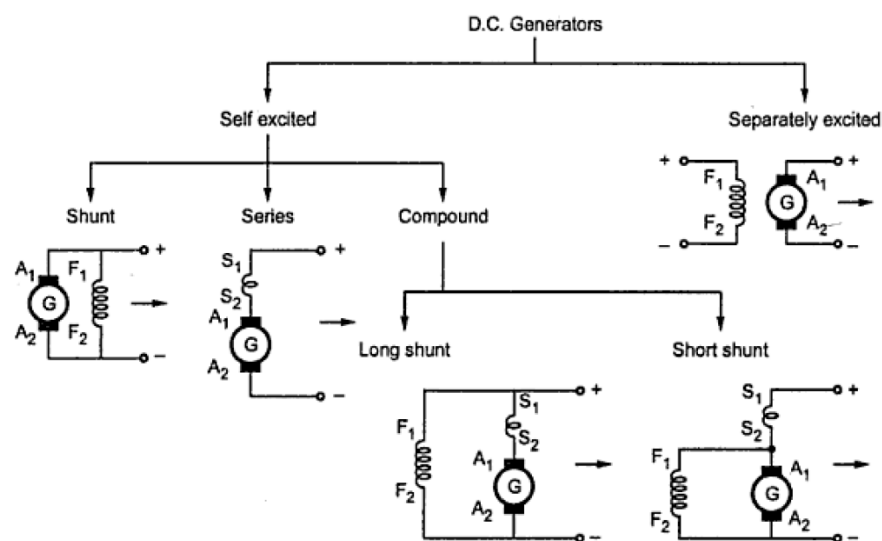


Figure 23: Types of DC generator connections.

16. Force on a Current-Carrying Conductor

When a current-carrying conductor is placed at right angles to a magnetic field, a mechanical force is experienced by the conductor. Its magnitude is $F = B I l$, and its direction is given by **Fleming's left-hand rule**.

Fleming's left-hand rule: stretch the thumb, forefinger and middle finger of the left hand mutually perpendicular. If the forefinger points along the magnetic field and the middle finger along the current, then the thumb gives the direction of the force on the conductor.

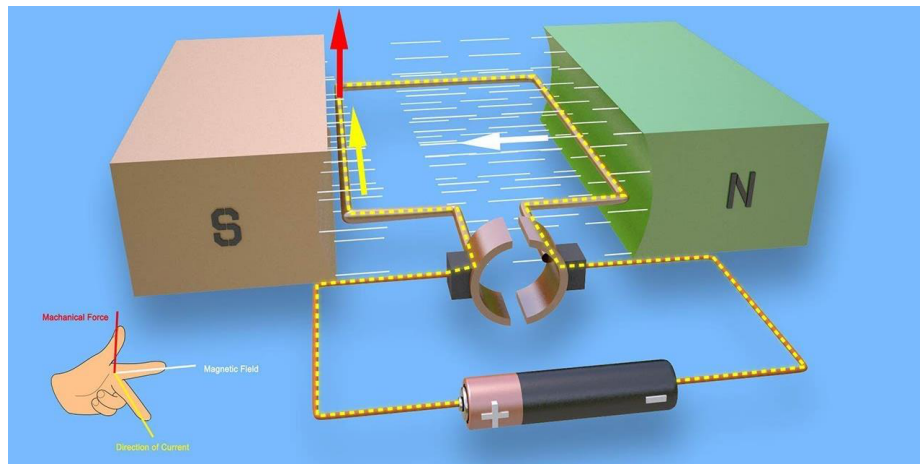


Figure 24: Force on a current-carrying conductor in a magnetic field.

17. DC Motor — Principle, Torque and Back EMF

When a DC machine is loaded as a motor, the rotor conductors carry current and lie in the magnetic field of the air-gap. Each conductor therefore experiences a force. Because the conductors lie at a common radius from the centre, a **torque** is produced and the rotor starts rotating.

17.1 Torque

Torque is the quantitative measure of the tendency of a force to cause or change a rotational motion — it is the moment of a force. For a DC motor:

$$T = 0.159 \phi Z I_a \text{ (P/A) (N·m)}$$

17.2 Back EMF (Counter EMF)

When the armature of a DC motor rotates, its conductors move through the magnetic field, so an emf is induced in them just as in a generator. By Lenz's law this induced emf opposes the applied voltage and is called the **back emf** or counter emf, E_b :

$$E_b = (\phi Z N P) / (60 A) \text{ volts}$$

The voltage equation of a DC motor is: $V = E_b + I_a R_a$

so that $E_b = V - I_a R_a$, where I_a is the armature current and R_a the armature resistance. The back emf thus regulates the current drawn by the motor.

17.3 Applications of DC Motors

- Shunt motors — machine tools, lathes, blowers, fans and general workshop drives.
- Series motors — cranes, hoists, electric traction, lifts (where a high starting torque is needed).
- Compound motors — rolling mills, presses, conveyors and shears.

18. Solved Numerical Examples

Example 1: Flux density

Problem: a magnetic pole face of 200 mm × 100 mm produces a total flux of 150 μWb. Find the flux density.

$$A = 0.2 \times 0.1 = 0.02 \text{ m}^2; \quad B = \phi/A = 150 \times 10^{-6} / 0.02 = 7.5 \times 10^{-3} \text{ T} = 7.5 \text{ mT}$$

Example 2: Radius of a pole face

Problem: the maximum working flux density of a lifting electromagnet is 1.8 T and the total flux produced is 353 mWb. Find the radius of the circular pole face.

$$A = \phi/B = 0.353/1.8 = 0.1961 \text{ m}^2; \quad r = \sqrt{(A/\pi)} = \sqrt{(0.1961/\pi)} = 0.25 \text{ m}$$

Example 3: Reluctance

Problem: find the reluctance of a metal piece of length 150 mm and cross-sectional area 1800 mm², if its relative permeability is 4000.

$$S = l/(\mu_0 \mu_r A) = 0.15/(4\pi \times 10^{-7} \times 4000 \times 1800 \times 10^{-6}) = 16579 \text{ AT/Wb}$$

Example 4: Turns from reluctance

Problem: a mild steel ring of radius 50 mm and cross-sectional area 400 mm² carries a flux of 0.1 mWb with a current of 0.5 A. If $\mu_r = 200$, find the reluctance and the number of turns.

$$l = 2\pi \times 0.05 = 0.3142 \text{ m}; \quad S = 0.3142/(4\pi \times 10^{-7} \times 200 \times 400 \times 10^{-6}) = 3.126 \times 10^6 \text{ AT/Wb}$$

$$\text{mmf} = \phi S = 0.1 \times 10^{-3} \times 3.126 \times 10^6 = 312.6 \text{ AT}; \quad N = \text{mmf}/I = 312.6/0.5 = 625 \text{ turns}$$

Example 5: Composite circuit with an air-gap

Problem: a cast steel core of area 2 cm² and mean length 25 cm has an air-gap of 1 mm and a coil of 5000 turns. If $\mu_r = 48000$, find the current needed for a flux density of 0.80 T in the air-gap.

$$\phi = B A = 0.8 \times 2 \times 10^{-4} = 1.6 \times 10^{-4} \text{ Wb}$$

$$S_{\text{steel}} = 0.25/(4\pi \times 10^{-7} \times 48000 \times 2 \times 10^{-4}) = 20723 \text{ AT/Wb}; \quad S_{\text{gap}} = 0.001/(4\pi \times 10^{-7} \times 2 \times 10^{-4}) = 3.979 \times 10^6 \text{ AT/Wb}$$

$$S = 20723 + 3.979 \times 10^6 = 4.0 \times 10^6 \text{ AT/Wb}; \quad \text{mmf} = \phi S = 640 \text{ AT}; \quad I = 640/5000 = 0.128 \text{ A}$$

Example 6: EMF equation

Problem: a 4-pole DC generator has 500 armature conductors, a flux of 0.02 Wb per pole and runs at 1200 rpm. Find the generated emf for (a) lap and (b) wave winding.

$$\text{(a) Lap: } A = P = 4, \quad E = (0.02 \times 500 \times 1200 \times 4)/(60 \times 4) = 200 \text{ V}$$

$$\text{(b) Wave: } A = 2, \quad E = (0.02 \times 500 \times 1200 \times 4)/(60 \times 2) = 400 \text{ V}$$

Example 7: Back emf and torque

Problem: a 220 V DC motor has an armature resistance of 0.5 Ω and draws an armature current of 20 A. Find the back emf.

$$E_b = V - I_a R_a = 220 - 20 \times 0.5 = 210 \text{ V}$$

19. Quick Reference — Key Formulae and Definitions

Quantity	Formula	Unit
Flux density	$B = \phi / A$	tesla (T)
Magnetomotive force	$F = N I$	ampere-turn (AT)
Magnetic field strength	$H = N I / l$	AT/m
Permeability	$\mu = \mu_0 \mu_r$; $B = \mu H$	H/m
Reluctance	$S = l / (\mu_0 \mu_r A)$	AT/Wb
Composite reluctance	$S = S_1 + S_2 + \dots$	AT/Wb
Faraday's law	$e = -N (d\phi/dt)$	volt
Dynamically induced emf	$e = B l v \sin \theta$	volt
EMF equation (generator)	$E = \phi Z N P / (60 A)$	volt
Back emf (motor)	$E_b = V - I_a R_a$	volt
Torque (motor)	$T = 0.159 \phi Z I_a (P/A)$	N·m

Key Definitions at a Glance

Term	Definition
Magnetic flux	Number of magnetic lines of force; unit weber (Wb).
Flux density	Flux per unit area perpendicular to it; unit tesla.
MMF	The cause of magnetic flux, $F = NI$; unit ampere-turn.
Reluctance	Opposition offered to flux by a magnetic circuit; unit AT/Wb.
Permeability	Property of a medium that decides the flux density for a given H.
B-H curve	Graph of flux density B against magnetising force H.
Hysteresis loop	B-H loop traced over a complete magnetising cycle; its area gives the hysteresis loss.
Faraday's law	Induced emf is proportional to the rate of change of flux linkages.
Lenz's law	The induced emf opposes the change of flux that produces it.
Commutator	A mechanical rectifier that converts the AC in the armature into DC in the external circuit.
Lap winding	$A = P$ parallel paths; for high current, low voltage machines.
Wave winding	$A = 2$ parallel paths; for high voltage, low current machines.
Back emf	The emf induced in a motor armature that opposes the applied voltage.